MAT 504

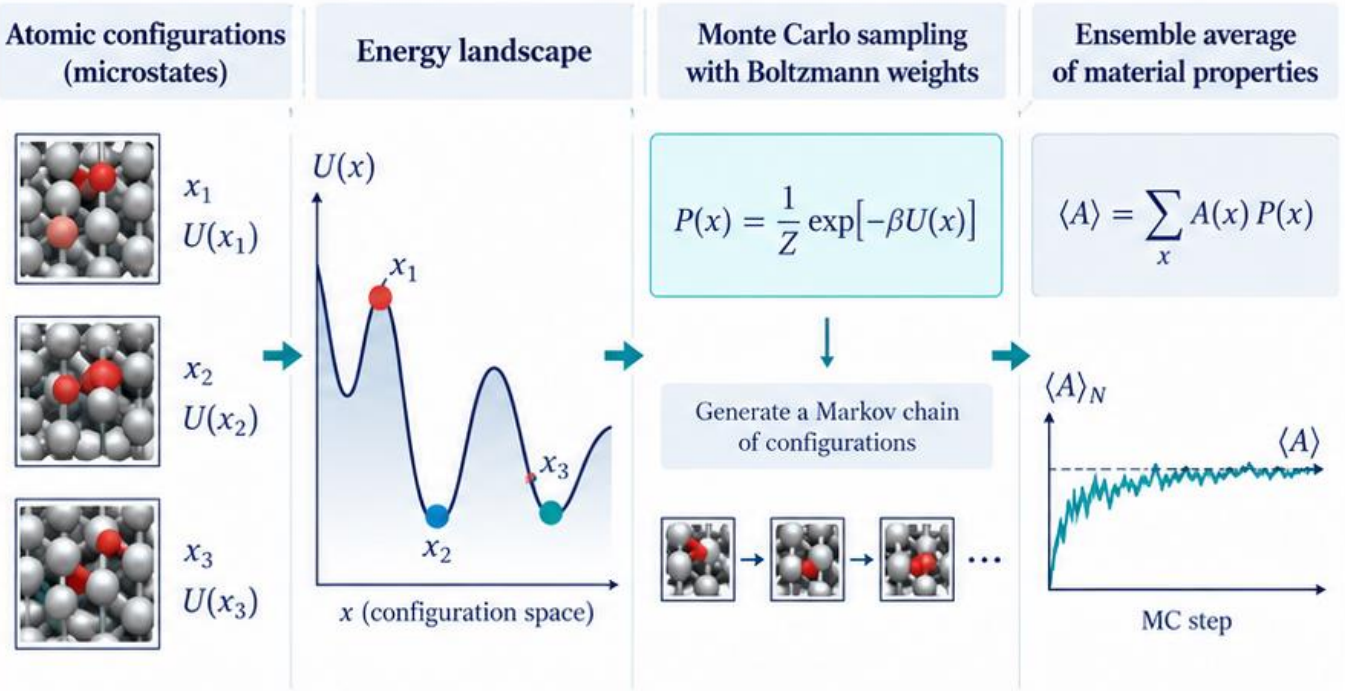
Computer Modelling in Materials Science

Week 2 — Statistical Mechanics and Monte Carlo Sampling

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Sabanci University

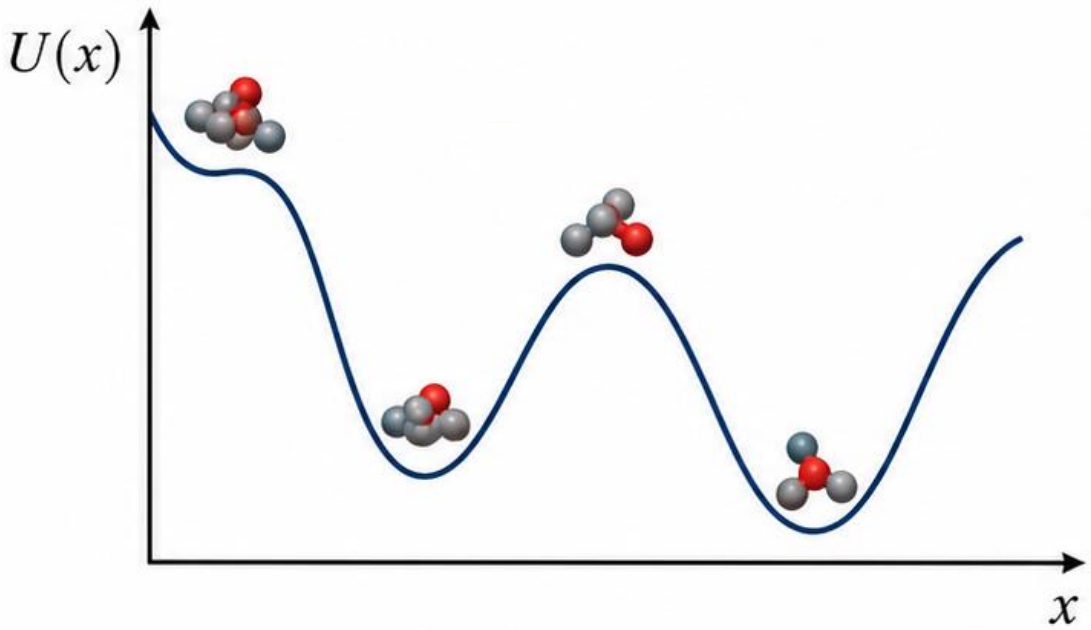
From atomic configurations to ensemble averages

Monte Carlo sampling generates representative configurations according to the **Boltzmann distribution.**



Statistical Mechanics and Monte Carlo Sampling

From an energy model to equilibrium averages



Boltzmann weights

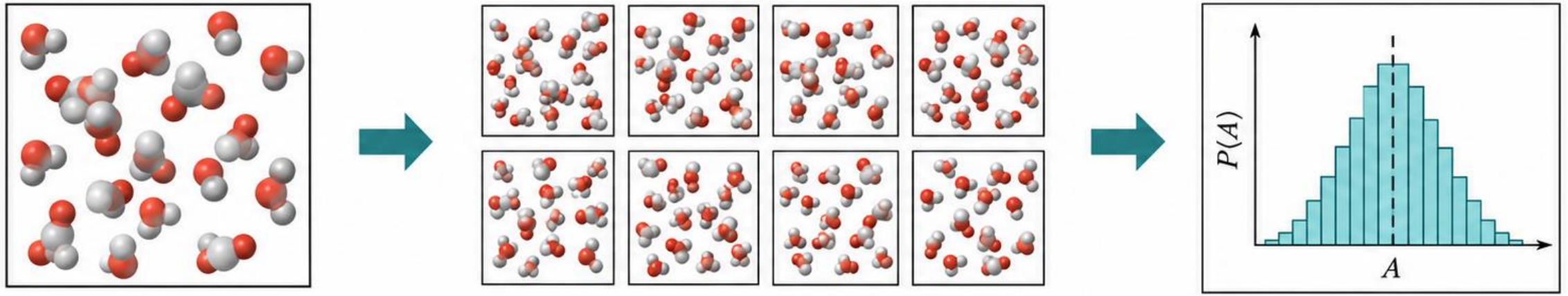
observable average

$$\langle A \rangle = \sum_x A(x) P(x)$$

Today: how to sample the configurations that matter.

From one configuration to a thermodynamic property

A single configuration is not the property.



one microstate x → many accessible configurations → estimate $\langle A \rangle$

sample

ensemble

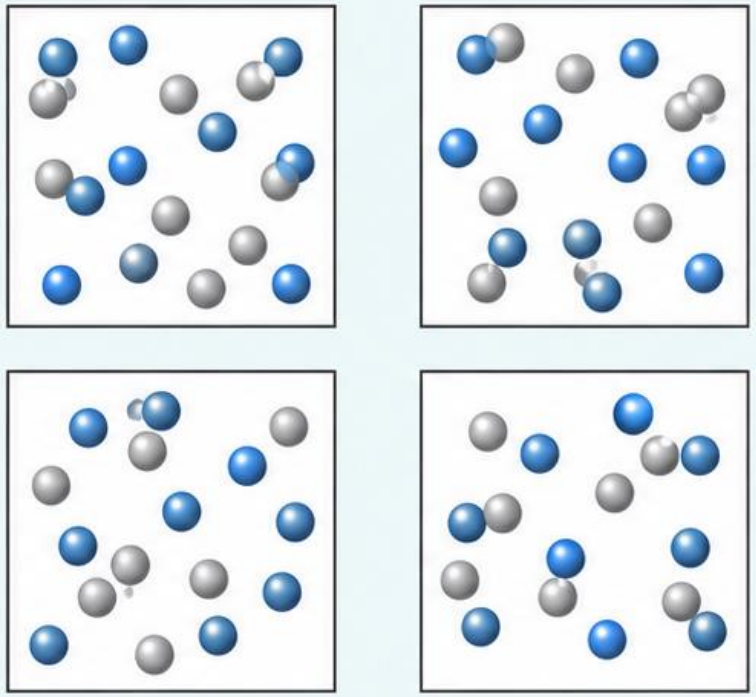
snapshot

- Properties are statistical.
- Configurations have different probabilities.
- Sampling replaces impossible enumeration.

We need rules for which configurations count, and how much.

Microstates, macrostates, and observables

Microstates



complete microscopic specification

many microstates can share the same macrostate

Macrostate

N, V, T

$\langle E \rangle$

density ρ

composition

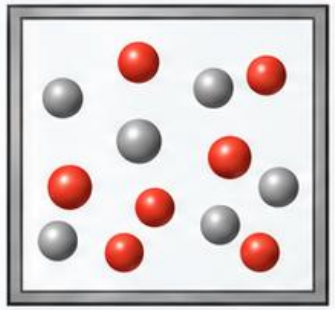
Observable: $A(\mathbf{x})$

$\langle A \rangle$ = average over sampled microstates

Statistical mechanics connects microscopic states to measurable macroscopic behavior.

What is an ensemble?

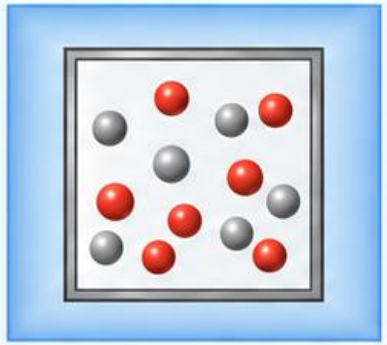
NVE
(microcanonical ensemble)



fixed
N, V, E

isolated system
(no exchange with surroundings)

NVT
(canonical ensemble)

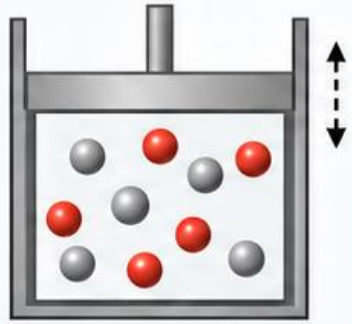


fixed
N, V, T

in contact with a heat bath
(exchanges energy)

Main focus today:
canonical ensemble (NVT)

NPT
(isothermal–isobaric ensemble)



fixed
N, P, T

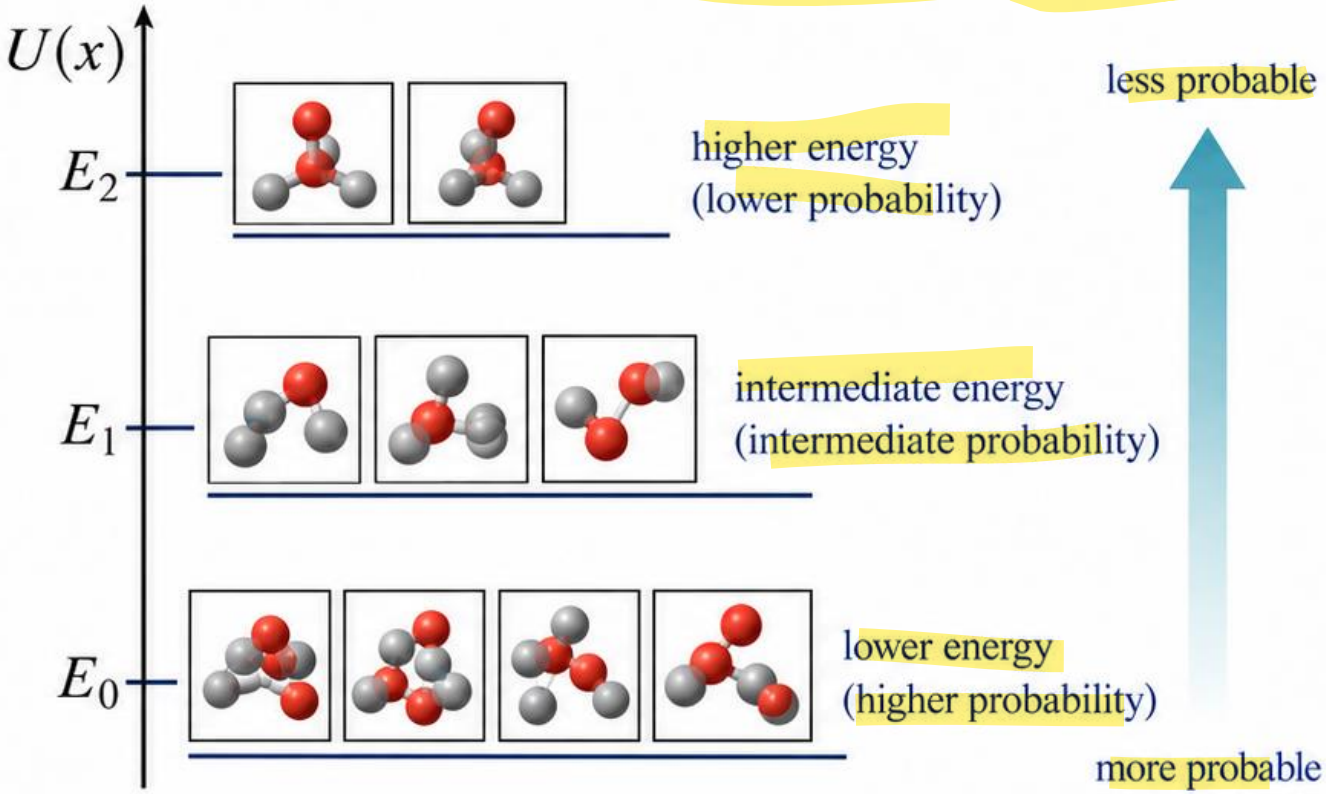
in contact with a heat and pressure bath (exchanges energy and volume)

- Ensemble = many microscopic realizations under the same macroscopic conditions.
- Different ensembles correspond to different constraints.
- Today we will sample the canonical ensemble.

An ensemble provides the statistical framework behind thermodynamic averages.

The canonical ensemble and the Boltzmann distribution

Not all microstates are equally probable in the canonical ensemble.



Boltzmann distribution

$$P(x) = \frac{\exp[-\beta U(x)]}{Z}$$

$$\beta = \frac{1}{k_B T}$$

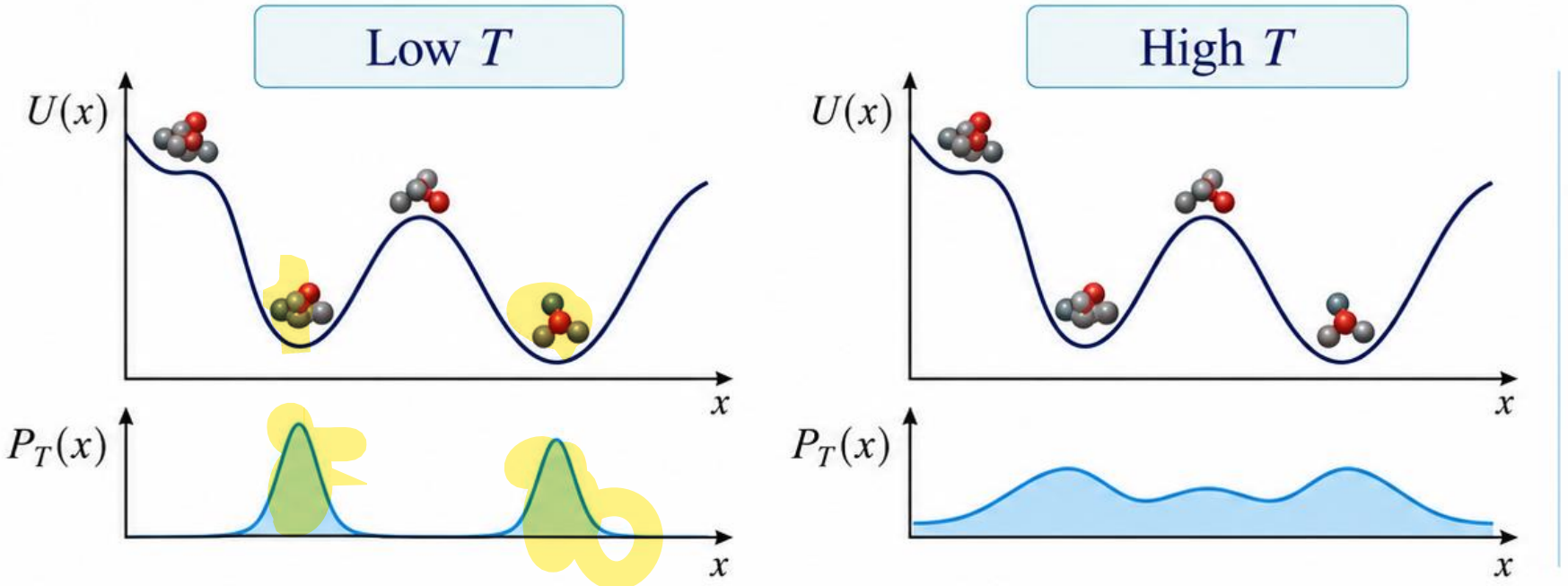
Z = partition function (normalization)

- Probability depends on energy and temperature.
- Lower-energy configurations carry larger statistical weight.
- At finite temperature, high-energy states are suppressed, not forbidden.

Equilibrium properties depend on weighted averages, not just the minimum-energy state.

What does temperature do?

For a fixed potential-energy model $U(x)$, temperature changes how configurations are statistically weighted.



Boltzmann distribution

$$P_T(x) \propto \exp\left[-\frac{U(x)}{k_B T}\right]$$

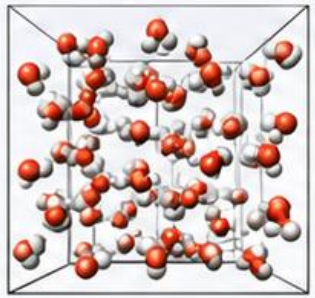
- Low T: probability is strongly concentrated in low-U regions.
- High T: higher-U configurations carry more statistical weight.
- Here $U(x)$ is fixed; temperature changes $P_T(x)$, not the potential-energy model.

Temperature changes Boltzmann weighting; the same ΔU matters differently at different T.

Thermodynamic properties are ensemble averages

A property (observable) is an average over many configurations, weighted by their equilibrium probabilities.

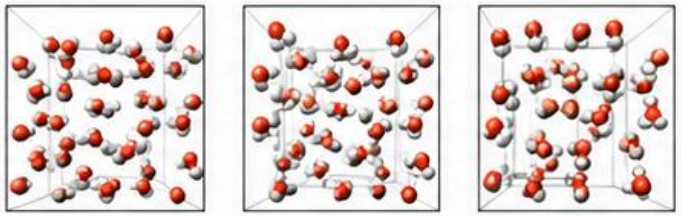
Configuration x



x

Potential energy
$U(x)$

Many accessible configurations



x_1 x_2 x_3

$U(x_1)$	$U(x_2)$	$U(x_3)$
$P(x_1)$	$P(x_2)$	$P(x_3)$

Ensemble average

$$\langle A \rangle = \sum_x A(x)P(x)$$

$$P(x) = \frac{1}{Z}\exp\left[-\beta U(x)\right]$$

$$Z = \sum_x \exp\left[-\beta U(x)\right]$$

$$\beta = \frac{1}{k_B T}$$

Examples of observables

Observable $A(x)$	Example / meaning
Potential energy $U(x)$	Total potential energy of the configuration
Total energy $E(x)$	Kinetic + potential energy
Particle density $\rho(r)$	Local number density at position r
Pressure P	Average force on simulation cell walls
Structural order	Radial distribution function $g(r)$
Number of H-bonds	Count of hydrogen bonds in the system

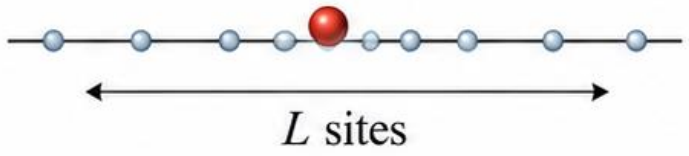
- Observables are quantities computed from a configuration (structural, energetic, or other physical properties).
- Each configuration has an equilibrium probability given by the Boltzmann distribution.
- Simulations estimate the ensemble average by sampling many configurations.

We do not compute a property from a single configuration, but from a representative sample.

Why can't we enumerate configuration space?

The number of possible configurations grows explosively with the number of degrees of freedom.

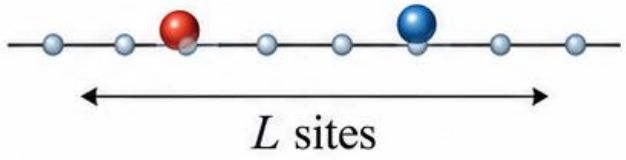
One particle on a 1D grid



$$\Omega = L$$

(L possible positions)

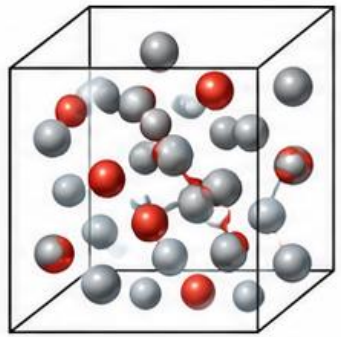
Two particles on a 1D grid



$$\Omega = L^2$$

(each particle can be at any of L sites)

Many-particle molecular system



N atoms

$3N$ coordinates

(x, y, z for each atom)

Configuration space dimension = $3N$

$$\Omega \sim L^{3N}$$

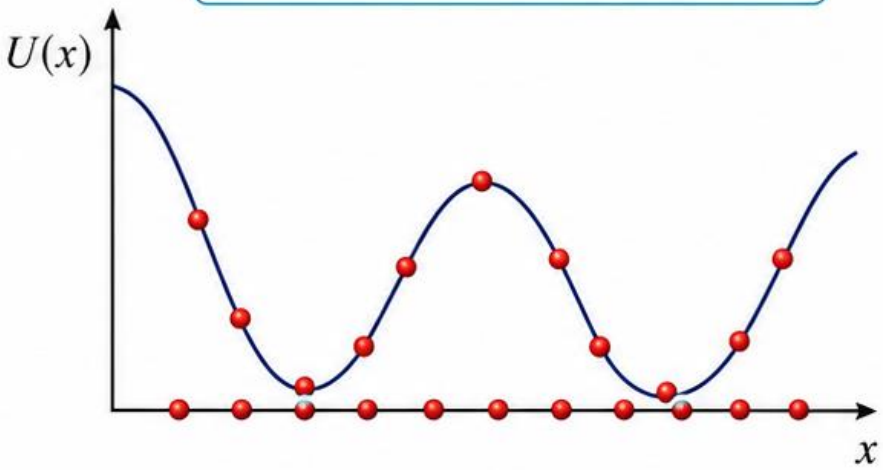
(grows exponentially with N)

Exhaustive enumartion is impossible, equilibrium properties must be estimated by sampling.

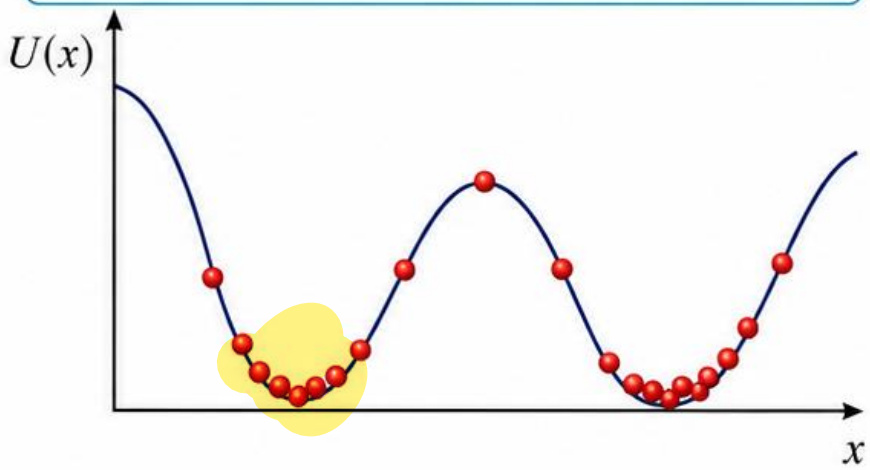
Importance sampling: spend effort where probability is high

We do not need every configuration equally; we need configurations with the correct statistical frequency.

Uniform sampling



Importance sampling (Boltzmann weighted)



Boltzmann distribution

$$P(x) \propto \exp\left[-\frac{U(x)}{k_B T}\right]$$

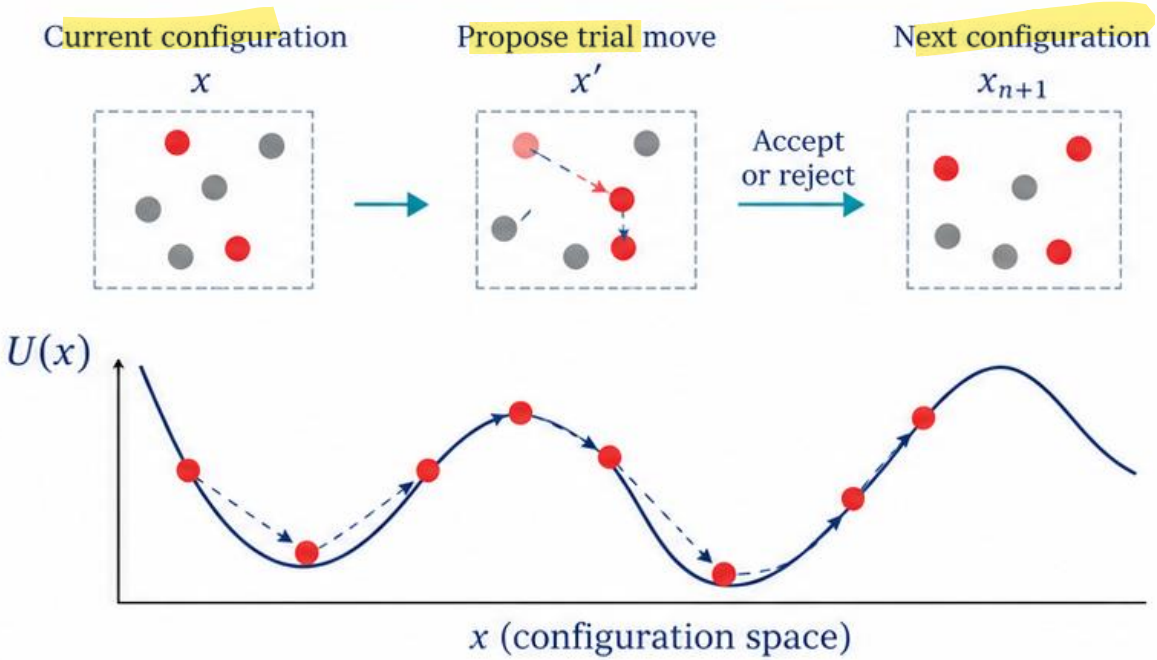
- Uniform sampling wastes effort in low-probability regions.
- Importance sampling concentrates effort where equilibrium probability is large.
- Correct averages are preserved because samples appear with the target distribution.

Importance sampling reduces wasted effort while preserving the correct equilibrium distribution.

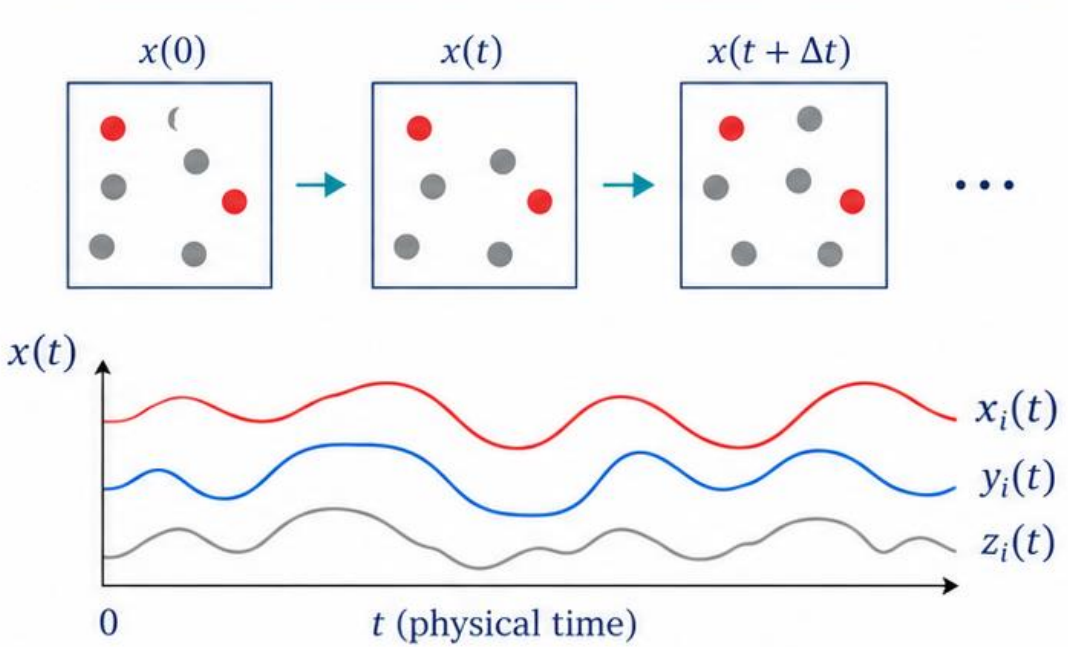
Monte Carlo is sampling, not dynamics

Monte Carlo generates configurations with the correct equilibrium distribution; MC step number is not physical time.

Monte Carlo sampling



Molecular dynamics

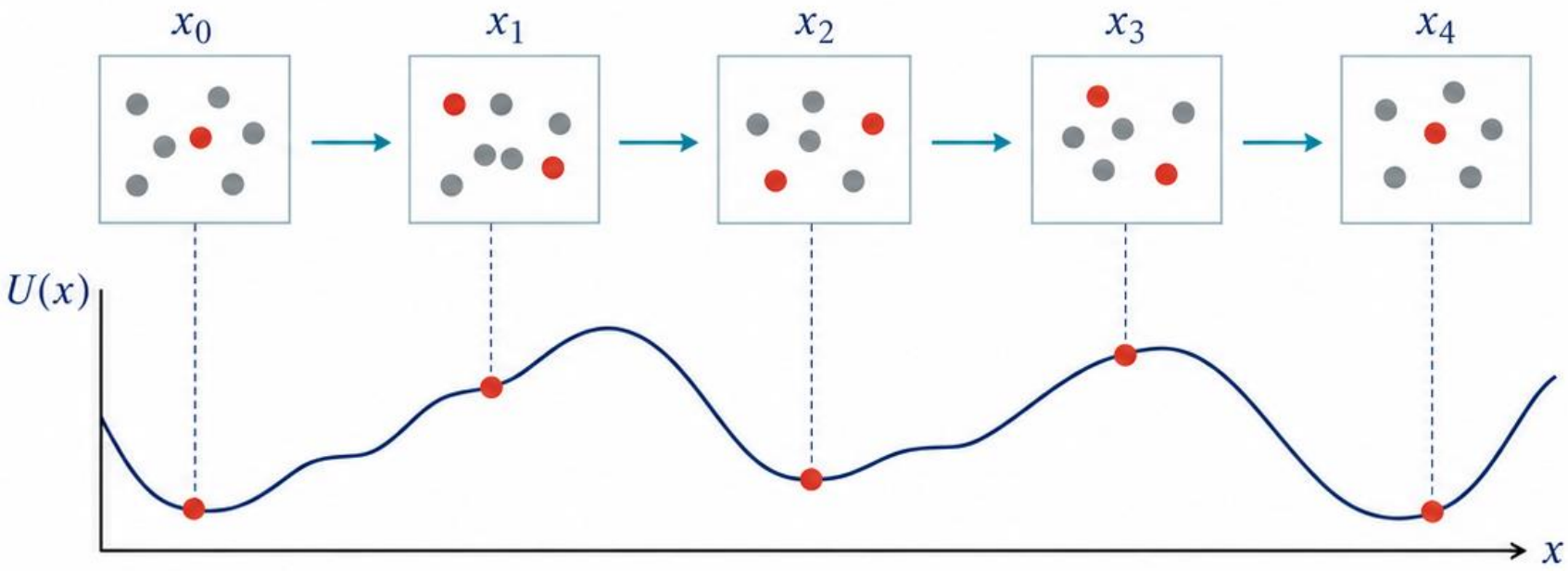


- MC proposes trial moves in configuration space.
- Acceptance/rejection is based on statistical weights, not equations of motion.
- MC estimates equilibrium averages, but it does not reproduce real molecular kinetics.

Monte Carlo is a sampling method for equilibrium properties, not a model of real-time dynamics.

Markov chains: generating configurations sequentially

Each new configuration is generated from the current one; the long-run distribution is designed to be the target ensemble.



Markov property

$$P(x_{n+1} \mid x_n, x_{n-1}, \ldots, x_0) = P(x_{n+1} \mid x_n)$$

Target stationary distribution

$$P(x) \propto \exp\left[-\beta U(x)\right]$$

- The next step depends on the current state, not the full history.
- Transition rules define a Markov chain in configuration space.
- We choose the rules so the stationary distribution is the Boltzmann distribution.

A Monte Carlo simulation constructs a chain whose long run behavior reproduces the target ensemble.

The Metropolis algorithm

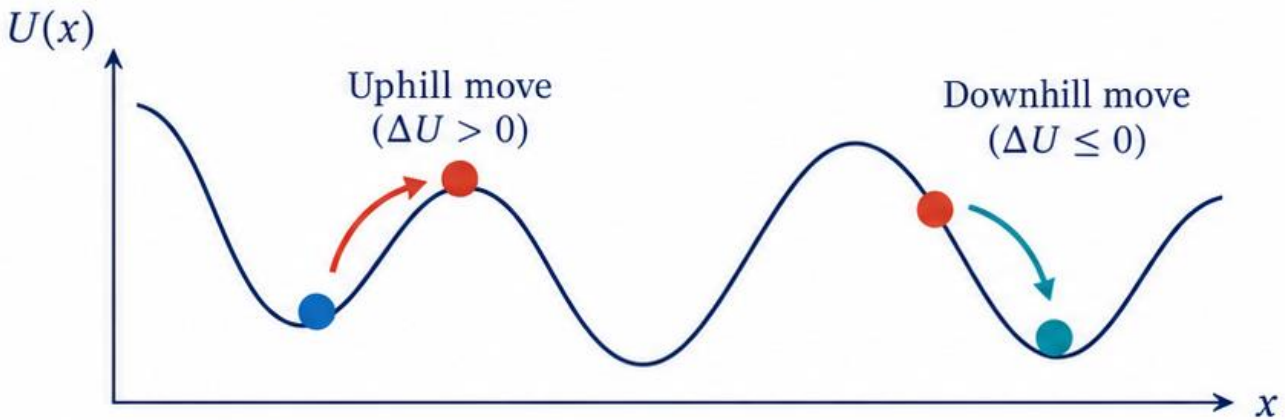
For symmetric trial moves, acceptance depends only on the energy change ΔU.

1. **Start from current configuration x.**
2. **Propose a trial configuration x'.**
3. **Compute** $\Delta U = U(x') - U(x)$.
4. **Accept with** $p_{\text{acc}} = \min[1, \exp(-\beta \Delta U)]$.



If $\Delta U \leq 0$: **accept always** ($p_{\text{acc}} = 1$)

If $\Delta U > 0$: **accept with probability** $\exp(-\beta \Delta U)$



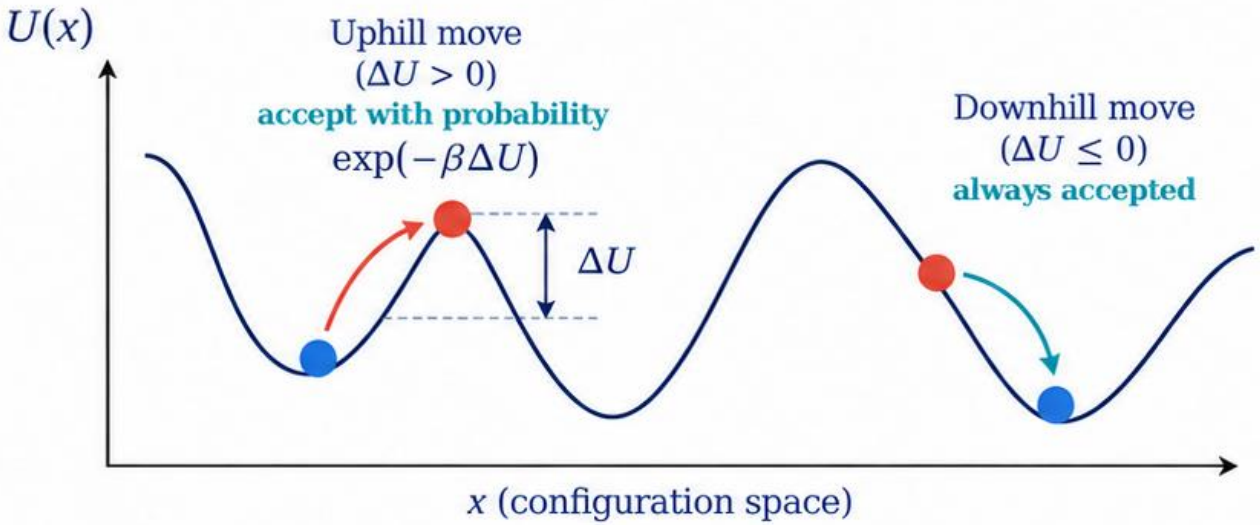
- Downhill moves are always accepted.
- Uphill moves are sometimes accepted.
- This avoids trapping and yields the canonical distribution.

Metropolis balances exploration with the correct equilibrium weighting.

Why do we sometimes accept uphill moves?

Downhill moves are always accepted; uphill moves are sometimes accepted so that Monte Carlo samples a finite-temperature equilibrium distribution rather than simply minimizing energy.

Energy landscape and trial moves



Temperature dependence of uphill acceptance

For $\Delta U > 0$: $p_{\text{acc}} = \exp(-\beta \Delta U)$

Low temperature
(small acceptance probability)



$T = 100$ K

$p_{\text{acc}} = \exp(-\beta \Delta U) \ll 1$

Uphill moves are rarely accepted.

High temperature
(larger acceptance probability)



$T = 1000$ K

$p_{\text{acc}} = \exp(-\beta \Delta U)$ (larger value)

Uphill moves are accepted more often.

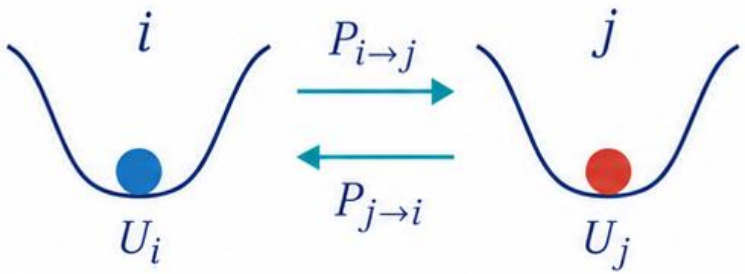
- Downhill moves are always accepted.
- Uphill moves are needed to avoid trapping in local minima.
- Finite-temperature sampling requires occasional uphill moves to reproduce the Boltzmann distribution.

Without uphill moves, the chain would simply descend in energy and get trapped.

Why does Metropolis sample the right distribution?

Detailed balance gives an intuitive route to the correct equilibrium distribution.

Two states and transitions



π_i = equilibrium probability of state i

$P_{i\to j}$ = probability of moving from i to j in one Monte Carlo step

$$\pi_i \, P_{i\to j} = \pi_j \, P_{j\to i}$$

At equilibrium, the probability flow from i to j is balanced by the reverse flow from j to i.

Transition probability = proposal × acceptance

The probability of moving from i to j is the product of a proposal probability and an acceptance probability:

$$P_{i\to j} = q_{i\to j}\, \alpha_{i\to j}$$

Symmetric proposal moves

$$q_{i\to j} = q_{j\to i}$$

Metropolis acceptance rule

$$\alpha_{i\to j} = \min\left[1, \exp(-\beta \Delta U)\right]$$

$$\Delta U = U_j - U_i$$

Why this gives Boltzmann sampling

For symmetric proposals, the ratio of transition probabilities is

$$\frac{P_{i\to j}}{P_{j\to i}} = \frac{\alpha_{i\to j}}{\alpha_{j\to i}} = \exp\left[-\beta(U_j - U_i)\right]$$

The equilibrium (Boltzmann) probability ratio is

$$\frac{\pi_j}{\pi_i} = \exp\left[-\beta(U_j - U_i)\right]$$

↓

The transition ratio matches the Boltzmann probability ratio.

Therefore detailed balance holds for

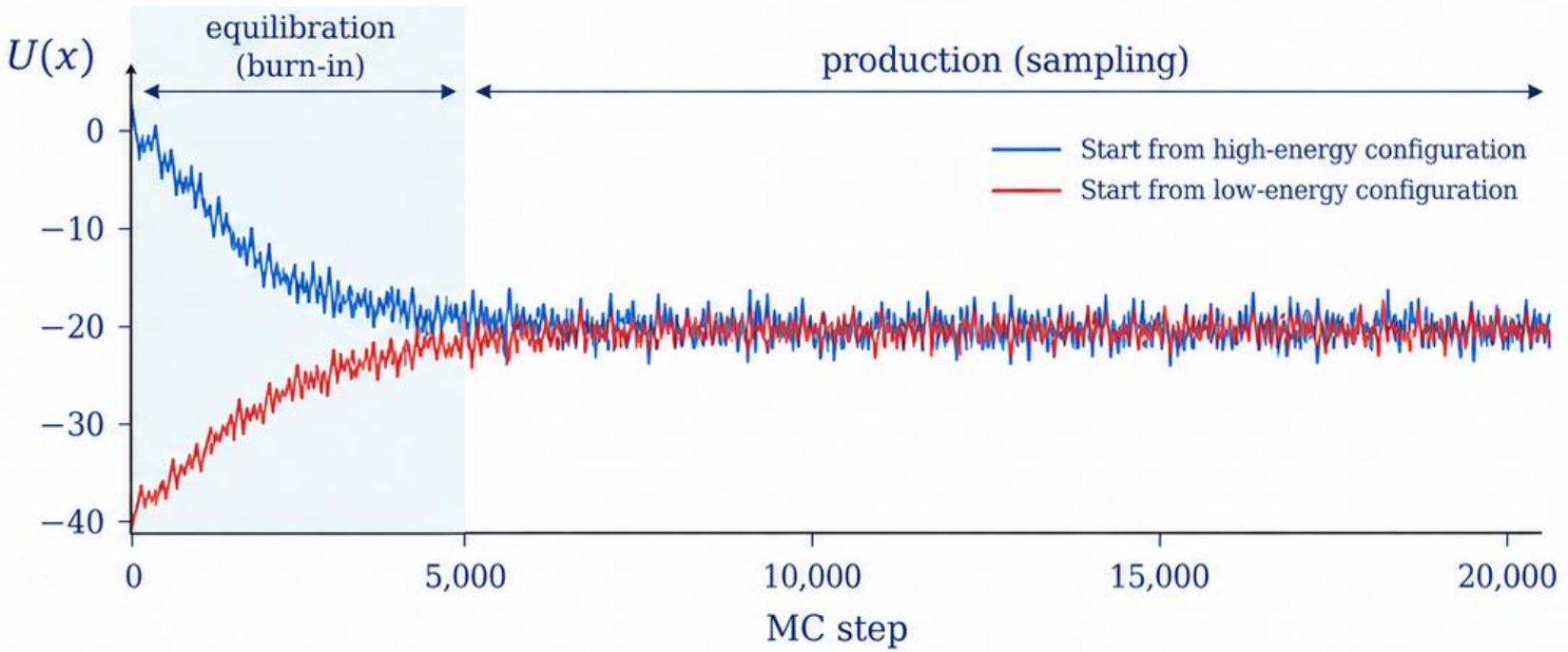
$$\pi_i \propto \exp(-\beta U_i).$$

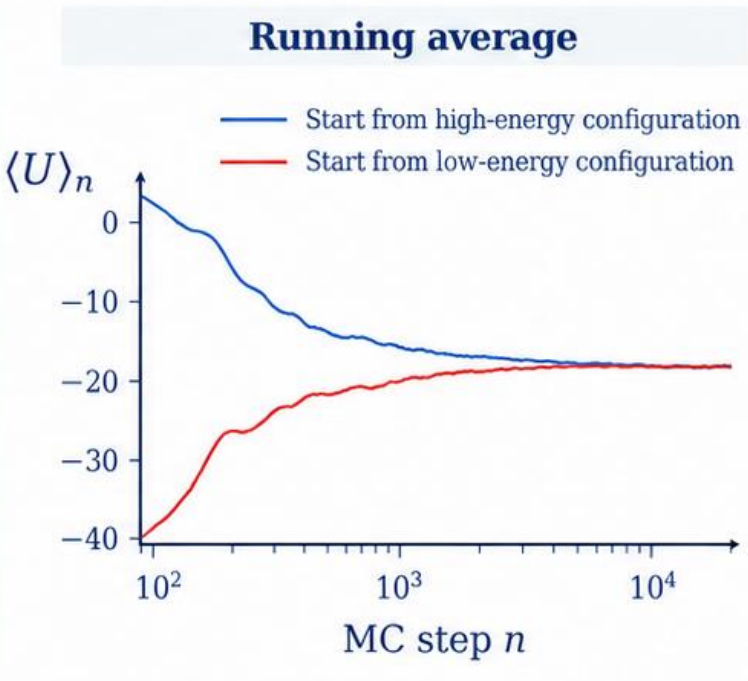
- Detailed balance is sufficient for the Boltzmann distribution to be a stationary distribution.
- Lower-energy moves are favored, while uphill moves are accepted with the exact probability needed to preserve Boltzmann weights.
- Metropolis works because its forward and reverse transitions have the correct equilibrium balance.

Metropolis samples correctly because its transition rules satisfy detailed balance with respect to the Boltzmann distribution.

Equilibration: when should we start measuring?

Initial configurations may not be representative; we should measure only after burn-in.





- Burn-in (equilibration): the chain relaxes from the initial configuration to the equilibrium distribution.
- Production sampling: measure observables only after burn-in.
- Check convergence: use different initial configurations (and/or multiple chains) to ensure consistent results.

Do not average until the chain has forgotten where it started.

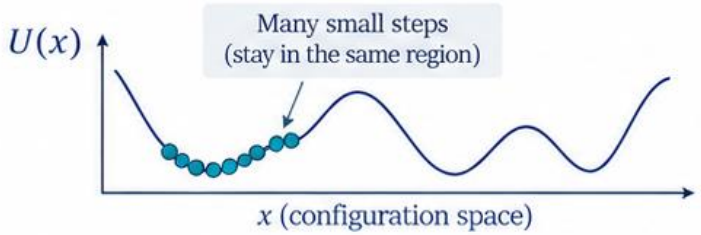
WEEK 2

Sampling efficiency: many steps ≠ many independent samples

The usefulness of a Monte Carlo simulation depends on how well it explores configuration space, not just the acceptance rate.

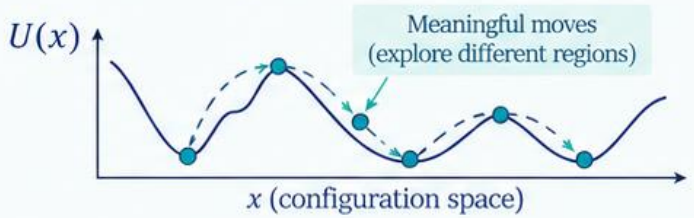
1. Moves too small: high acceptance, slow exploration

- Trial moves are very small.
- Most moves are accepted.
- The chain changes only a little each step and moves slowly through configuration space.
- Consecutive samples are strongly correlated.



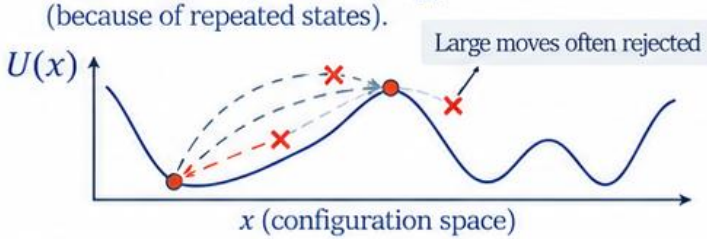
2. Balanced moves: good exploration

- Trial moves are of a moderate size.
- Moves are accepted often enough, but also change the configuration meaningfully.
- The chain can move between regions and explore configuration space efficiently.
- Consecutive samples become less correlated.



3. Moves too large: many rejections, poor exploration

- Trial moves are very large.
- Many proposed moves land in high-energy regions and are rejected.
- The chain often stays in the same configuration for many steps.
- Consecutive samples are strongly correlated (because of repeated states).



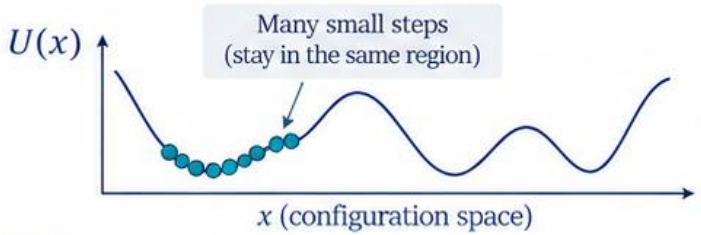
WEEK 2

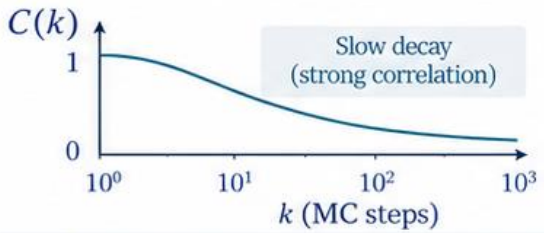
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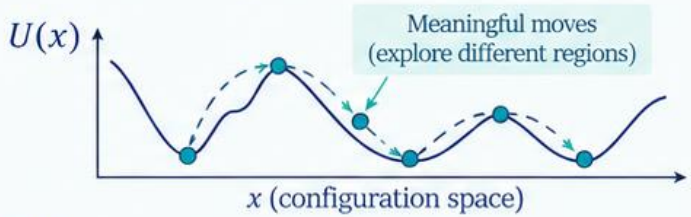


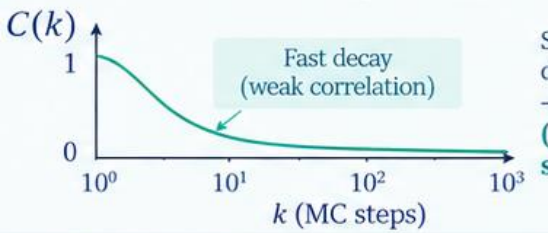
Samples remain similar for many steps – **few independent samples.**

Acceptance rate: high
Exploration: poor (slow)
Samples: strongly correlated

2. Balanced moves: good exploration

- Trial moves are of a moderate size.
- Moves are accepted often enough, but also change the configuration meaningfully.
- The chain can move between regions and explore configuration space efficiently.
- Consecutive samples become less correlated.



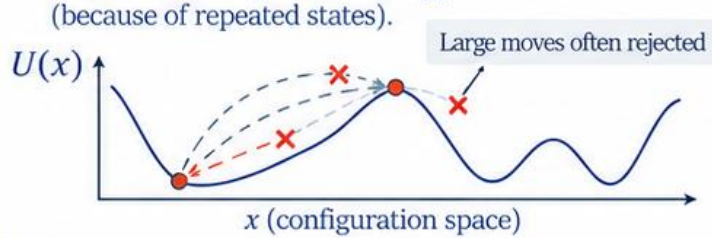


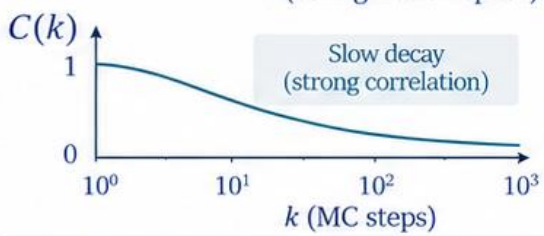
Samples become different quickly – **more useful (independent) samples.**

Acceptance rate: moderate
Exploration: good (across barriers)
Samples: more independent

3. Moves too large: many rejections, poor exploration

- Trial moves are very large.
- Many proposed moves land in high-energy regions and are rejected.
- The chain often stays in the same configuration for many steps.
- Consecutive samples are strongly correlated (because of repeated states).

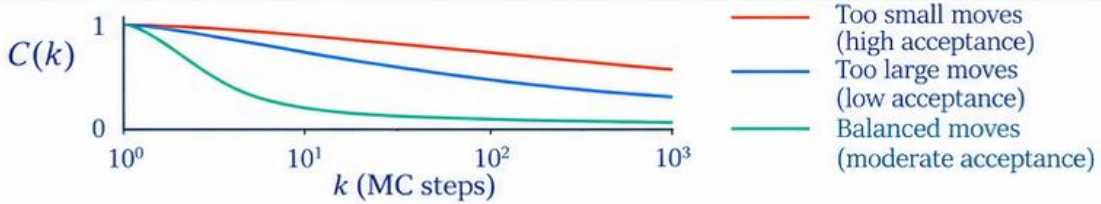




Chain often stays in the same state – **few independent samples.**

Acceptance rate: low
Exploration: poor (many rejections)
Samples: strongly correlated

Autocorrelation comparison



Key takeaways

- Many Monte Carlo steps do not mean many independent samples.
- Consecutive samples are correlated; the autocorrelation function $C(k)$ measures how similar samples are k steps apart.
- Both very small and very large moves can give poor sampling (slow decorrelation).
- A balanced move size gives a reasonable acceptance rate and more efficient exploration.
- Effective sample size depends on the autocorrelation time, not the total number of steps.

WEEK 2

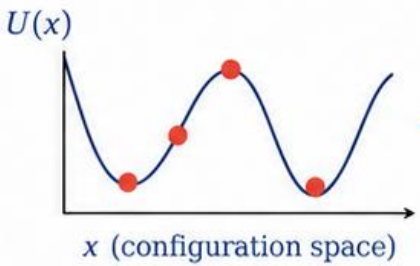
What should I check before trusting a Monte Carlo result?

Acceptance rate alone is not enough; correctness depends on the target distribution, equilibration, exploration, convergence, and reproducibility.

1. Correct target distribution

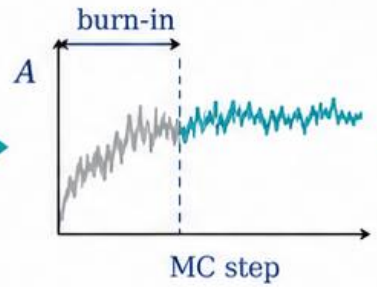
Are the configurations distributed as

$P(x) \propto \exp[-\beta U(x)]$?



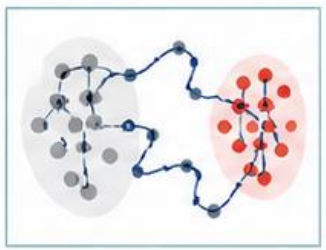
2. Equilibration / burn-in

Discard an initial transient before collecting data.



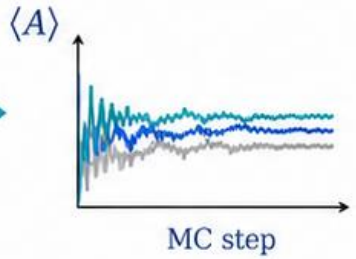
3. Adequate exploration (mixing)

Ensure the chain visits relevant regions of configuration space (avoids trapping).



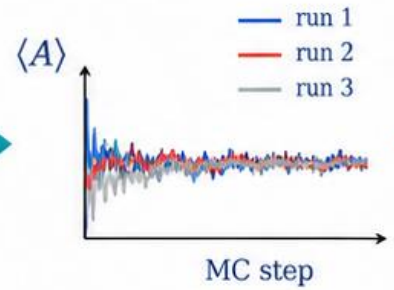
4. Convergence / stable averages

Check that running averages are stable and independent of further sampling.



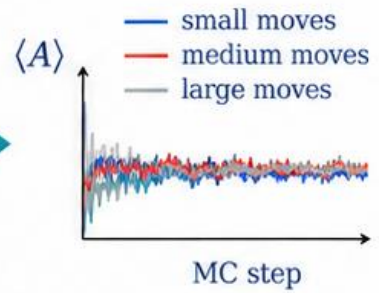
5. Reproducibility from independent starts

Run multiple independent chains with different initial configurations (random seeds).



6. Sensitivity to move size / proposals

Check that results are robust to the choice of move size or proposal distribution.



Acceptance rate alone is not proof of correctness.

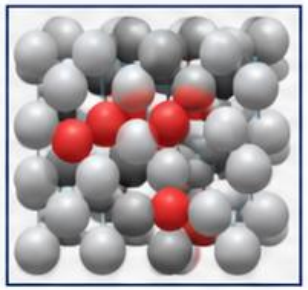
- A reasonable acceptance rate (e.g. 20–60%) is often useful for efficient sampling, but it does not guarantee the correct distribution, good exploration, or convergence.
- A high or low acceptance rate can still give biased results if the chain is not equilibrated or well mixed.
- Always check the diagnostic points above.

Trust comes from diagnostics, not from a single acceptance-rate number.

Takeaways: from energy model to equilibrium sampling

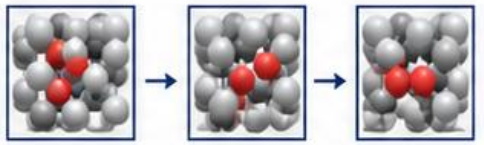
Monte Carlo turns an energy model into representative configurations and ensemble averages.

Energy model	Boltzmann distribution	Monte Carlo sampling	Ensemble average
Assign an energy to each configuration.	Equilibrium probability of a configuration.	Generate configurations by a Markov chain (e.g. Metropolis).	Compute material properties as equilibrium averages.



$U(x)$

$$P(x) \propto \exp\left[-\beta U(x)\right]$$



x_0 x_1 x_2

$$\langle A \rangle = \sum_x A(x) P(x)$$

- **Next:** implement Metropolis sampling, examine acceptance, equilibration, and averaging.

The goal is not a trajectory in time, but the correct equilibrium average.